

Васильев В.М., РКН^о1, ИУ5Ц-81Б, вариант N^о26

```
# Импорт необходимых библиотек
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_wine

# Загрузка датасета Wine
wine = load_wine()
X, y = wine.data, wine.target
feature_names = wine.feature_names
target_names = wine.target_names

# Создание DataFrame
df = pd.DataFrame(X, columns=feature_names)
df['target'] = y
df['wine_class'] = df['target'].map({i: target_names[i] for i in
range(3)})

# Проверка на пропущенные значения
print("Количество пропущенных значений:")
print(df.isnull().sum())

# Удаление строк с пропусками (если бы они были)
# df = df.dropna()
```

```
Количество пропущенных значений:
alcohol          0
malic_acid       0
ash              0
alcalinity_of_ash 0
magnesium        0
total_phenols    0
flavanoids       0
nonflavanoid_phenols 0
proanthocyanins  0
color_intensity  0
hue              0
od280/od315_of_diluted_wines 0
proline          0
target           0
wine_class       0
dtype: int64
```

```
# Анализ структуры данных
print("\nИнформация о данных:")
print(df.info())
print("\nСтатистика данных:")
print(df.describe())
```

Информация о данных:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 178 entries, 0 to 177

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	alcohol	178 non-null	float64
1	malic_acid	178 non-null	float64
2	ash	178 non-null	float64
3	alcalinity_of_ash	178 non-null	float64
4	magnesium	178 non-null	float64
5	total_phenols	178 non-null	float64
6	flavanoids	178 non-null	float64
7	nonflavanoid_phenols	178 non-null	float64
8	proanthocyanins	178 non-null	float64
9	color_intensity	178 non-null	float64
10	hue	178 non-null	float64
11	od280/od315_of_diluted_wines	178 non-null	float64
12	proline	178 non-null	float64
13	target	178 non-null	int64
14	wine_class	178 non-null	object

dtypes: float64(13), int64(1), object(1)

memory usage: 21.0+ KB

None

Статистика данных:

	alcohol	malic_acid	ash	alcalinity_of_ash
count	178.000000	178.000000	178.000000	178.000000
mean	13.000618	2.336348	2.366517	19.494944
std	0.811827	1.117146	0.274344	3.339564
min	11.030000	0.740000	1.360000	10.600000
25%	12.362500	1.602500	2.210000	17.200000
50%	13.050000	1.865000	2.360000	19.500000
75%	13.677500	3.082500	2.557500	21.500000
max	14.830000	5.800000	3.230000	30.000000

162.000000

	total_phenols	flavanoids	nonflavanoid_phenols
proanthocyanins \			
count	178.000000	178.000000	178.000000
178.000000			
mean	2.295112	2.029270	0.361854
1.590899			
std	0.625851	0.998859	0.124453
0.572359			
min	0.980000	0.340000	0.130000
0.410000			
25%	1.742500	1.205000	0.270000
1.250000			
50%	2.355000	2.135000	0.340000
1.555000			
75%	2.800000	2.875000	0.437500
1.950000			
max	3.880000	5.080000	0.660000
3.580000			

	color_intensity	hue	od280/od315_of_diluted_wines
proline \			
count	178.000000	178.000000	178.000000
178.000000			
mean	5.058090	0.957449	2.611685
746.893258			
std	2.318286	0.228572	0.709990
314.907474			
min	1.280000	0.480000	1.270000
278.000000			
25%	3.220000	0.782500	1.937500
500.500000			
50%	4.690000	0.965000	2.780000
673.500000			
75%	6.200000	1.120000	3.170000
985.000000			
max	13.000000	1.710000	4.000000
1680.000000			

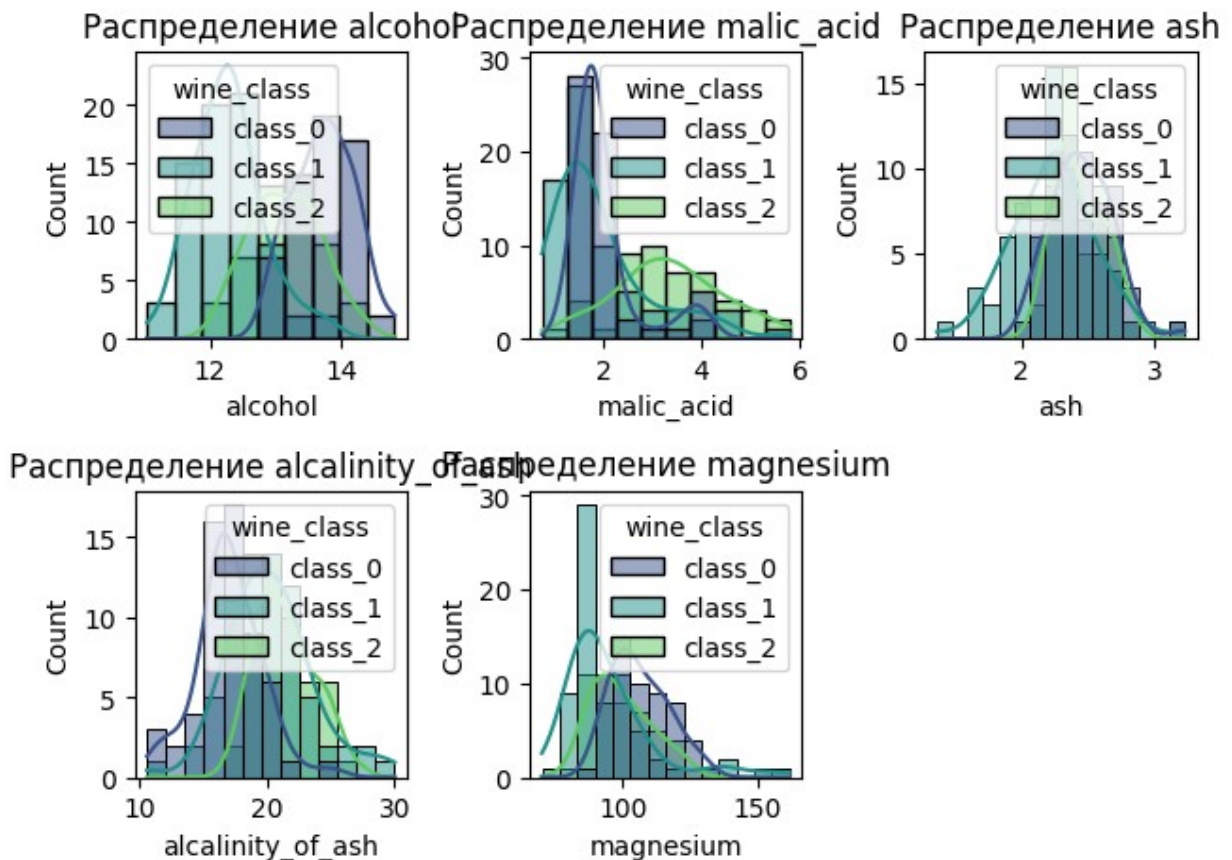
	target
count	178.000000
mean	0.938202
std	0.775035
min	0.000000
25%	0.000000
50%	1.000000
75%	2.000000
max	2.000000

```
# Визуализация
plt.figure(figsize=(15, 10))

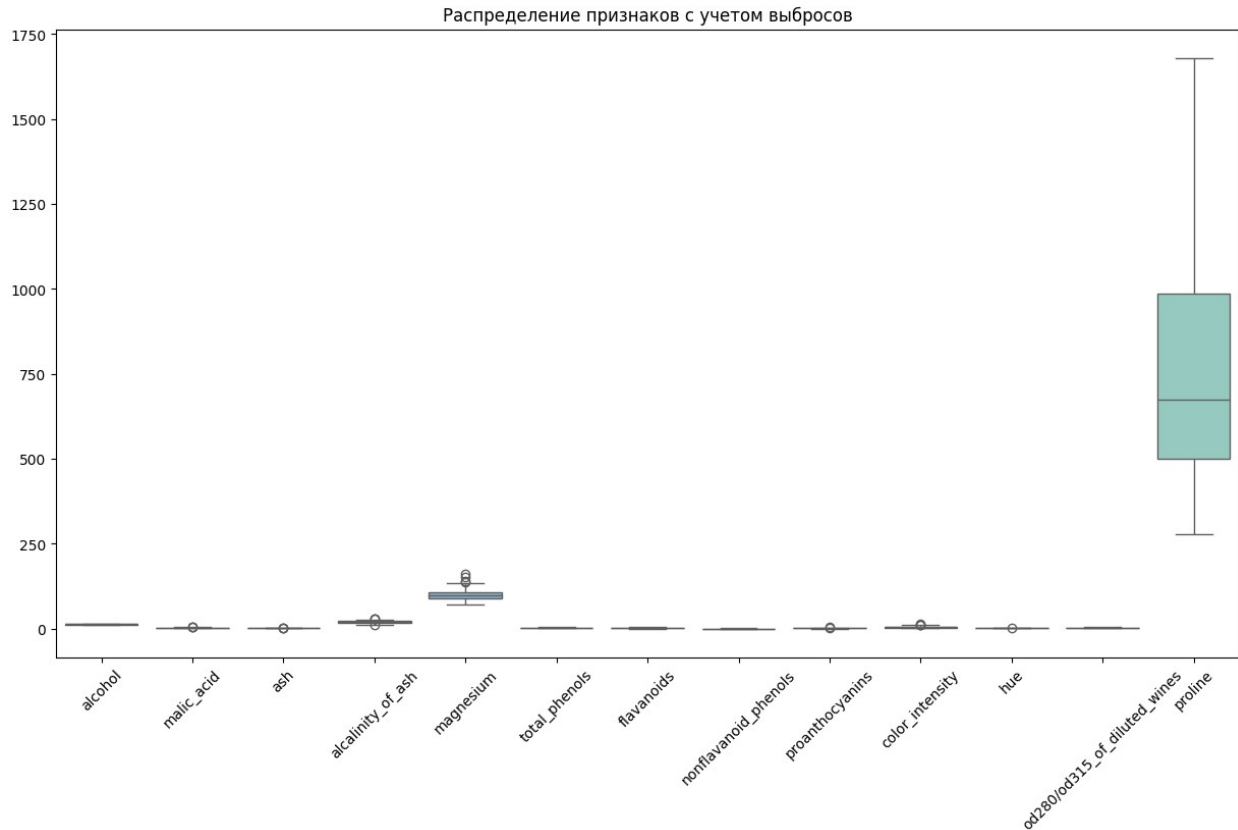
<Figure size 1500x1000 with 0 Axes>

<Figure size 1500x1000 with 0 Axes>

# 1. Гистограммы распределения признаков
for i, feature in enumerate(feature_names[:5], 1):
    plt.subplot(2, 3, i)
    sns.histplot(data=df, x=feature, kde=True, hue='wine_class',
palette='viridis')
    plt.title(f'Распределение {feature}')
plt.tight_layout()
plt.show()
```

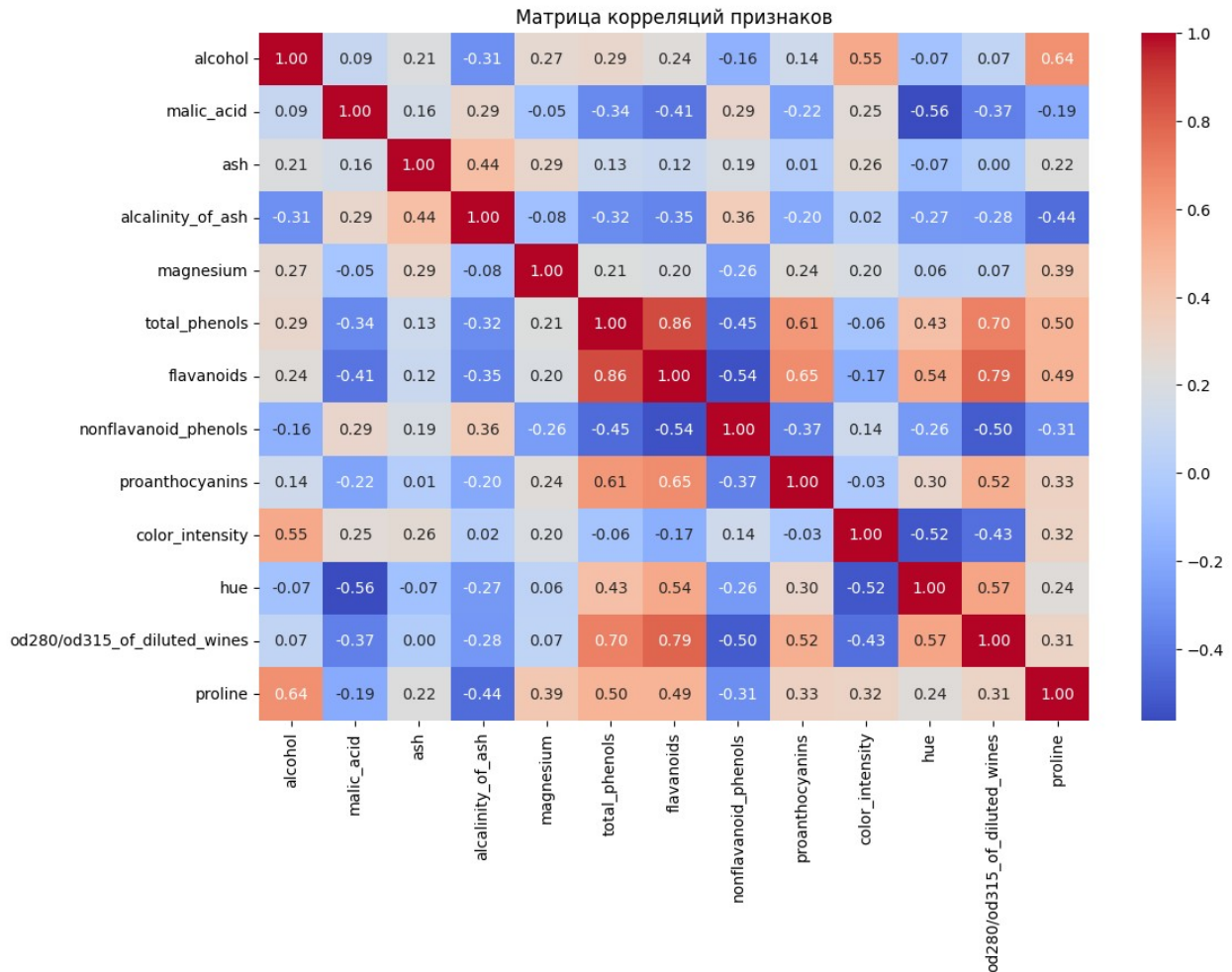


```
# 2. Boxplot для анализа выбросов
plt.figure(figsize=(15, 8))
sns.boxplot(data=df.drop(columns=['target', 'wine_class']),
palette='Set3')
plt.xticks(rotation=45)
plt.title('Распределение признаков с учетом выбросов')
plt.show()
```



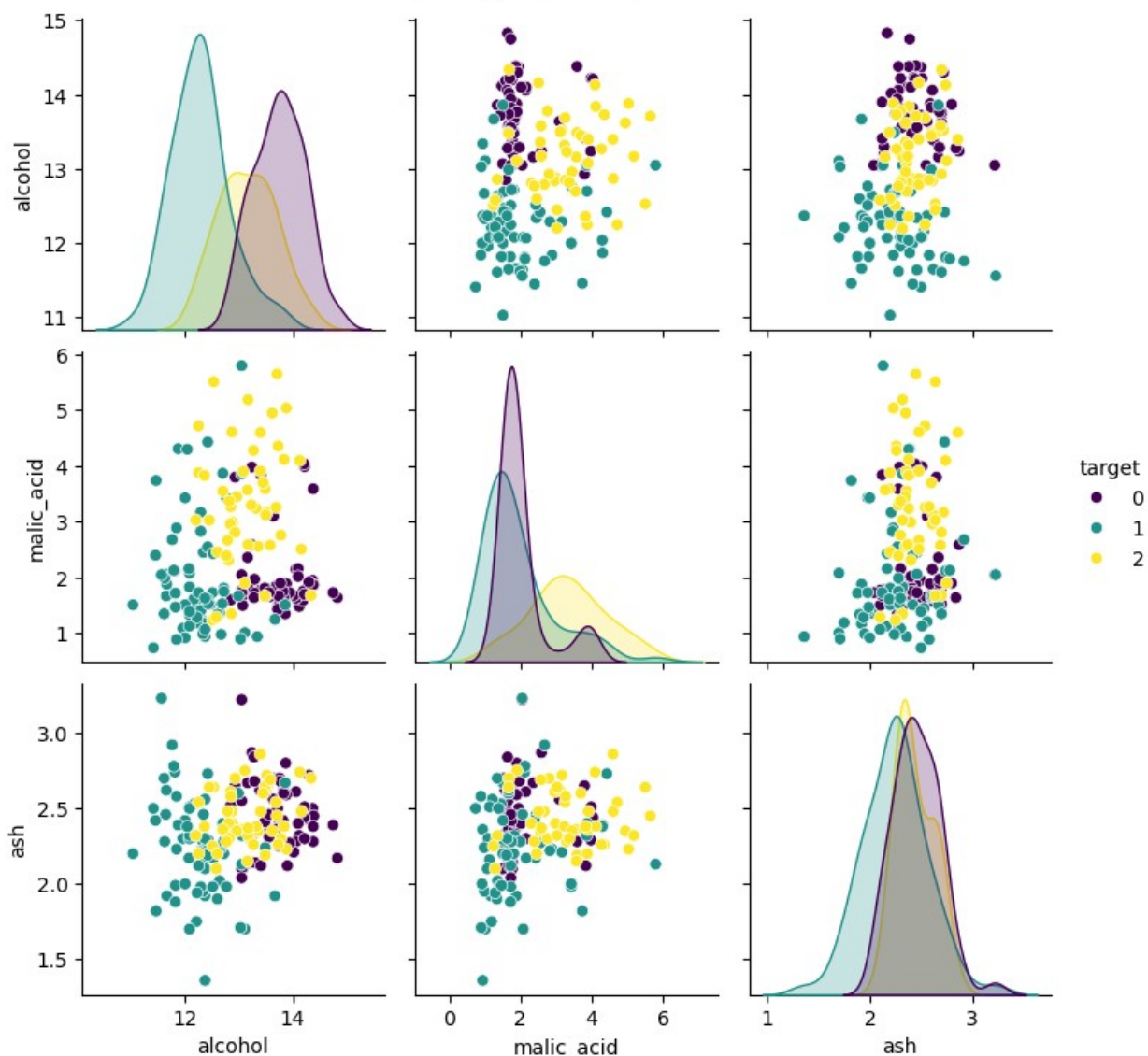
3. Тепловая карта корреляций

```
plt.figure(figsize=(12, 8))
corr_matrix = df.drop(columns=['target', 'wine_class']).corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Матрица корреляций признаков')
plt.show()
```



```
# 4. Scatter plot для пар признаков
sns.pairplot(df[['alcohol', 'malic_acid', 'ash', 'target']],
hue='target', palette='viridis')
plt.suptitle('Парные диаграммы рассеяния', y=1.02)
plt.show()
```

Парные диаграммы рассеяния



5. Распределение классов

```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='wine_class', palette='viridis')
plt.title('Распределение классов вин')
plt.xlabel('Класс')
plt.ylabel('Количество')
plt.show()
```

C:\Temp\ipykernel_1296\2885467239.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.countplot(data=df, x='wine_class', palette='viridis')
```

